Code-Layout, Readability and Reusability Template

Date	15 March 2026
Team ID	Batch 2023-27 | ACOE
Project Name	Rising Waters – ML-Based Flood Prediction System
Maximum Marks	3 Marks

Project Directory Structure

File / Directory	**Purpose**	**Author**
app.py	Main Flask controller — all routes, session handling	Venkatesh B.
model/train_model.py	Complete ML pipeline: preprocessing, training, save	Gode S R D.
model/generate_dataset.py	Synthetic 5,000-row meteorological dataset generator	Shivatmika G.
model/floods.save	Serialized XGBoost model (Joblib) — 96.55% accuracy	All Members
model/scaler.save	Fitted StandardScaler object (Joblib) for normalization	All Members
utils/preprocessing.py	Validation and scaling helper functions	Archana D.
templates/base.html	Shared Jinja2 layout: navigation, footer, CSS imports	Bavisetty G. K.
templates/result_flood.html	Dedicated Flood Chance result page	Bavisetty G. K.
templates/result_noflood.html	Dedicated No Flood result page	Bavisetty G. K.

File / Directory	Purpose	Author
static/css/style.css	Premium UI design	All Members
requirements.txt	Python dependencies	Venkatesh B.
Dockerfile	Docker container configuration	Venkatesh B.

Code Quality Standards Applied

S.No	Standard	Implementation
1	Module Docstrings	Every Python file begins with a purpose docstring
2	Single Responsibility	Each function performs one task, max 40 lines
3	Constants	FEATURE_NAMES, MODEL_PATH defined at module level
4	Template Inheritance	All HTML pages extend base.html — zero repetition
5	CSS Variables	--primary-color, --accent-color used for theming
6	HTTP Status Codes	All Flask routes return appropriate status codes
7	Input Validation	Server-side validation before model inference